

Technology Stack Justification

Smart-Lender Project Documentation Suite

Project Name	Smart Lender - AI-Powered Loan Approval Prediction System
Document Type	Technology Stack Justification
Project Phase	Phase 2: Requirement Analysis
Submitted By	Team Leader: Palli Dharani Team Members: Yarabarla Manidweep Saikiran Kudipudi Sita Rama Raju Datla Thota Leela Sai Krishna
Organization / College	Vishnu Institute of Technology
Status / Version	Approved / v1.0 (Exact Replica Format)

Technology Stack Matrix

Architecture Layer	Chosen Component Solution	Business & Technical Justification
Web Back-end	Flask web framework	Lightweight, native Python script launcher suitable for running serialized models
Machine Learning	XGBoost (Extreme Gradient Boosting)	Outperforms conventional models (Decision Tree, KNN, RF) on tabular credit records
Data Preprocessing	Scikit-Learn, Pandas, NumPy	Industry-standard utilities for cleaning, balancing (SMOTE) and scaling dataset features
Model Serializer	Joblib	Provides efficient compression for large Scikit-Learn pipelines and numpy matrices
Presentation Web UI	HTML5, CSS3, Jinja2	Enables clean CSS interfaces and dynamic template parameter injection